

Software Quality Factors Analysis Proposal

The chosen application is a Kotlin mobile application. Its primary function is tracking daily medication that the user has to take. It can be connected to a smartwatch to receive sensor gathered data like heart rate. The application sends notifications when sensor values go above a threshold and when medication needs to be taken.

The Quality Model I've chosen is the **ISO 25010** from 2011.

Func. Suit.	Perform. Efficiency	Compatib.	Usability	Reliability	Security	Maintainability	Portability
Func. Compl.	Time behav	Co-existen.	Appropriat.	Maturity	Confid.	Modularity	Adaptab.
Func. Corect.	Resource util.	Interoperability	Learna-bility	Availability	Integrity	Reusability	Instability
Func. Appro.	Capacity		Operability	Fault tolerance	Non-repudiance	Analysa-bility	Replace-bility
			User error prot.	Recoverability	Authenticity	Modifiability	
			UI Asethet.		Accountab	Testability	
			Accesibil.				

I've chosen 5 subfactors:

- **Resource Utilisation**
- **Recoverability**
- **Confidentiality**
- **Maintainability**
- **Installability**

This is how I plan to measure them:

Resource Utilization - it's important because the app is using background services to receive data from smart watches and to send notifications and this shouldn't be very consuming on a mobile device, otherwise it will drain the battery very fast.

Resource Utilization measurement:

- I will manually test on different physical devices using phones profiles (statistics collected by the phone to measure an apps performance)
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Recoverability - it's important because the app integrates data received from a smartwatch and it is important that the system can recover in case something happens to the watch (goes out of range, runs out of battery, etc.), or if the phone restarts

Recoverability measurement:

- mean time to repair (average time it takes to restore the system after a failure) measured heuristically
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Confidentiality - it's important because it uses sensitive user data (medical data)

Confidentiality measurement:

- static analysis by myself using a checklist (is there authentication, encrypted data transfer, etc.)
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Installability - it's important because being a mobile application it needs to run on different devices with different Android versions.

Installability measurement:

- Manual installation on different devices (with different Android versions)
 - And/Or the usage of Firebase test lab (emulated devices)
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Maintainability - it's important because all applications need to be maintained one way or another, this one in particular because it has the potential of being extended with more functionalities.

Maintainability measurement:

- Static analysis using automated tools in order to determine the Technical Debt and other issues.